

A Counterexample to Conjecture 00000002040: Largest Sum-Free Subsets of $[n]$ Have Density $\rightarrow 1/2$, Not $1/4$

Feiyu*

Claude (Opus 5.5)[†]

2026-10-02

Abstract

Conjecture 00000002040 asserts that the density of the largest 3-independent (sum-free) subset of $[n] = \{1, \dots, n\}$ tends to a limit that is at most, and in fact exactly, $1/4$. The largest such subset has exactly $\lceil n/2 \rceil$ elements, so the density tends to $1/2$. The odd numbers show that $1/2$ is attained, and a short reflection argument shows it cannot be beaten. Everything is formalized in Lean 4 against Mathlib.

1 The conjecture as stated

The original text of conjecture 00000002040 reads:

Definition: A 3-independent set has no solutions of $x + y = z$. Conjecture: The density limit of maximal 3-independent sets in $[n]$ exists and is at most $1/4$ (with correction 0); the limit is exactly $1/4$ (attained by the natural ternary cascade construction).

2 Reading of the statement

A set A of positive integers is *3-independent* if there are no $x, y, z \in A$ (not necessarily distinct) with $x + y = z$; such sets are usually called sum-free. Let

$$M(n) = \max\{|A| : A \subseteq [n] \text{ 3-independent}\}, \quad d(n) = M(n)/n.$$

“Maximal” refers to the largest such sets; the Chinese version of the statement (in the conjecture file) says so explicitly, speaking of the largest 3-independent set in $[n]$. The conjecture asserts that $d(n)$ converges, that its limit is at most $1/4$, and that the limit is exactly $1/4$. The parenthetical remarks (“with correction 0”, “attained by the natural ternary cascade construction”) add no mathematical content that we need.

3 Result

Theorem 1. *For every $n \geq 0$, $M(n) = \lceil n/2 \rceil = \lfloor (n+1)/2 \rfloor$. Consequently $d(n) \rightarrow \frac{1}{2}$, so $d(n)$ has no limit $\leq \frac{1}{4}$, and Conjecture 00000002040 is false.*

*Independent researcher.

[†]Anthropic.

4 Proof

Lemma 2 (lower bound). *The set of odd numbers in $[n]$ is 3-independent and has $\lfloor (n+1)/2 \rfloor$ elements.*

Proof. A sum of two odd numbers is even, hence not odd. The odd numbers in $[n]$ are $2k+1$ for $0 \leq k < \lfloor (n+1)/2 \rfloor$. $\square$

Lemma 3 (upper bound). *Every 3-independent $A \subseteq [n]$ has $|A| \leq \lfloor (n+1)/2 \rfloor$.*

Proof. We may assume $A \neq \emptyset$; let $m = \max A$. For $a \in A$ with $a < m$, the number $m - a$ is not in A , since otherwise $a + (m - a) = m$ would be a solution inside A . So A and $B = \{m - a : a \in A, a < m\}$ are disjoint. Both lie in $[m]$, and $|B| = |A| - 1$. Hence $2|A| - 1 \leq m \leq n$. $\square$

Proof of Theorem 1. Lemmas 2 and 3 give $M(n) = \lfloor (n+1)/2 \rfloor$. For $n \geq 1$ this gives $\frac{1}{2} \leq d(n) \leq \frac{1}{2} + \frac{1}{2n}$, so $d(n) \rightarrow \frac{1}{2}$. A convergent sequence has a unique limit, and $\frac{1}{2} > \frac{1}{4}$. $\square$

5 Formalization

The accompanying Lean 4 project (`lean/Submission/Basic.lean`) states the conjecture and proves its negation against Mathlib.

- `IsThreeIndependent A` — no $x, y, z \in A$ with $x + y = z$; `maxSize n` — the largest size of a 3-independent subset of `Finset.Icc 1 n`, a supremum over its powerset; `density n = maxSize n / n` in $\mathbb{R}$.
- `ConjectureHolds` — there is $L \leq \frac{1}{4}$ with `density  $\rightarrow L$` , and `density  $\rightarrow \frac{1}{4}$` .
- `odds, card_odds, isThreeIndependent_odds` — Lemma 2.
- `card_le_of_isThreeIndependent` — Lemma 3; `maxSize_eq` — $M(n) = \lfloor (n+1)/2 \rfloor$.
- `half_le_density, density_le, tendsto_density` — $d(n) \rightarrow \frac{1}{2}$.
- `conjecture_00000002040_false` — $\neg \text{ConjectureHolds}$, by uniqueness of limits.

The toolchain is `leanprover/lean4:v4.33.1`, Mathlib is pinned to `v4.33.1`, and `lake build` succeeds. The project contains no `sorry`, no `admit` and no `native_decide`, and introduces no axioms:

```
#print axioms conjecture_00000002040_false
```

reports exactly `propext`, `Classical.choice` and `Quot.sound`.

6 Remarks

The value $\lceil n/2 \rceil$ is classical. Besides the odd numbers, the upper half $\{\lfloor n/2 \rfloor + 1, \dots, n\}$ is also extremal. The constant $1/4$ does not arise for sum-free subsets of $[n]$. For comparison, in the cyclic group $\mathbb{Z}/p\mathbb{Z}$ (p prime) the largest sum-free subsets have $\lfloor (p+1)/3 \rfloor$ elements, density about $1/3$.